Bromination – Green Chemistry: A Written Lab Report

Christopher Naser

Department of Chemistry, Wayne State University

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Dr. Sean P. Hickey

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Abstract

In this experiment, my objective was to convert acetanilide to p-bromoacetanilide through a green chemistry experiment with reagents in situ, and I expected to retrieve accurate H-NMR and IR spectra results alongside a narrow and reasonable melting point range with a high percent yield. In general, acetanilide was placed in situ with several chemicals to induce an EAS reaction to place a bromine in the para position, which was then recrystallized and sampled to graphical utility. It was determined that a 19.87% percent recovery and 9.886% percent yield were retrieved with a melting point range of 163 – 166 °C and graphical H-NMR and IR spectra results. Finally, it was concluded that the product received was pure, which was supported by H-NMR and IR spectra results, but the sample had a low percent yield.

Introduction

This report encloses the bromination of acetanilide to p-bromoacetanilide and the importance of green chemistry when conducting laboratory experiments. Acetanilide consists of a benzene ring with an amide substitute, which is known to be an ortho/para director. An ortho/para director are substitutes on benzene rings that guide the nucleophilic substitution of an electrophile, which is known as Electrophilic Aromatic Substitution (EAS). Ortho/para directors direct the substitution of electrophiles because of the partial negative charges to those carbon positions on the benzene ring via resonance, which is induced from the electron donating group (the director). Electron donating groups are benzene substitutes that donate electron density to the aromatic ring, which tends to place such partial negative charges onto the ortho/para positions that induce a nucleophilic character. Then, by substitution (SN1 like), a bromine will be added to the benzene ring, most likely at the para position because it is slightly more stable than the ortho position due to less steric hinderance and better molecular geometry (*figure 1*).

Furthermore, to prevent the use of toxic chemicals like Br_2, which can be used for the bromination of acetanilide, this experiment focuses on a green alternative to this reaction. Instead, the generated electrophilic bromine is synthesized in situ by reacting NaClO and NaBr in the presence of acetic acid. This reaction will form BrOH, which is a better alternative than Br_2.

The purpose of this experiment was to convert acetanilide to p-bromoacetanilide through an EAS reaction. In short, during the experiment, ethanol, acetic acid, acetanilide, sodium bromide, and sodium hypochlorite were placed in a flask indulged into an ice bath. After a while, sodium thiosulfate and sodium hydroxide were used to quench the excess bromine and stop the experiment. Finally, the product was vacuumed and recrystallized to obtain a pure product, which was sampled under an IR and H-NMR spectrometer alongside a melting point apparatus.

For this experiment, I expect/hypothesize to retrieve appropriate, graphical H-NMR and IR spectra alongside an accurate and narrow melting point range that identifies a pure sample of p-bromoacetanlide. Also, I hypothesize a high percent yield for this experiment, too.

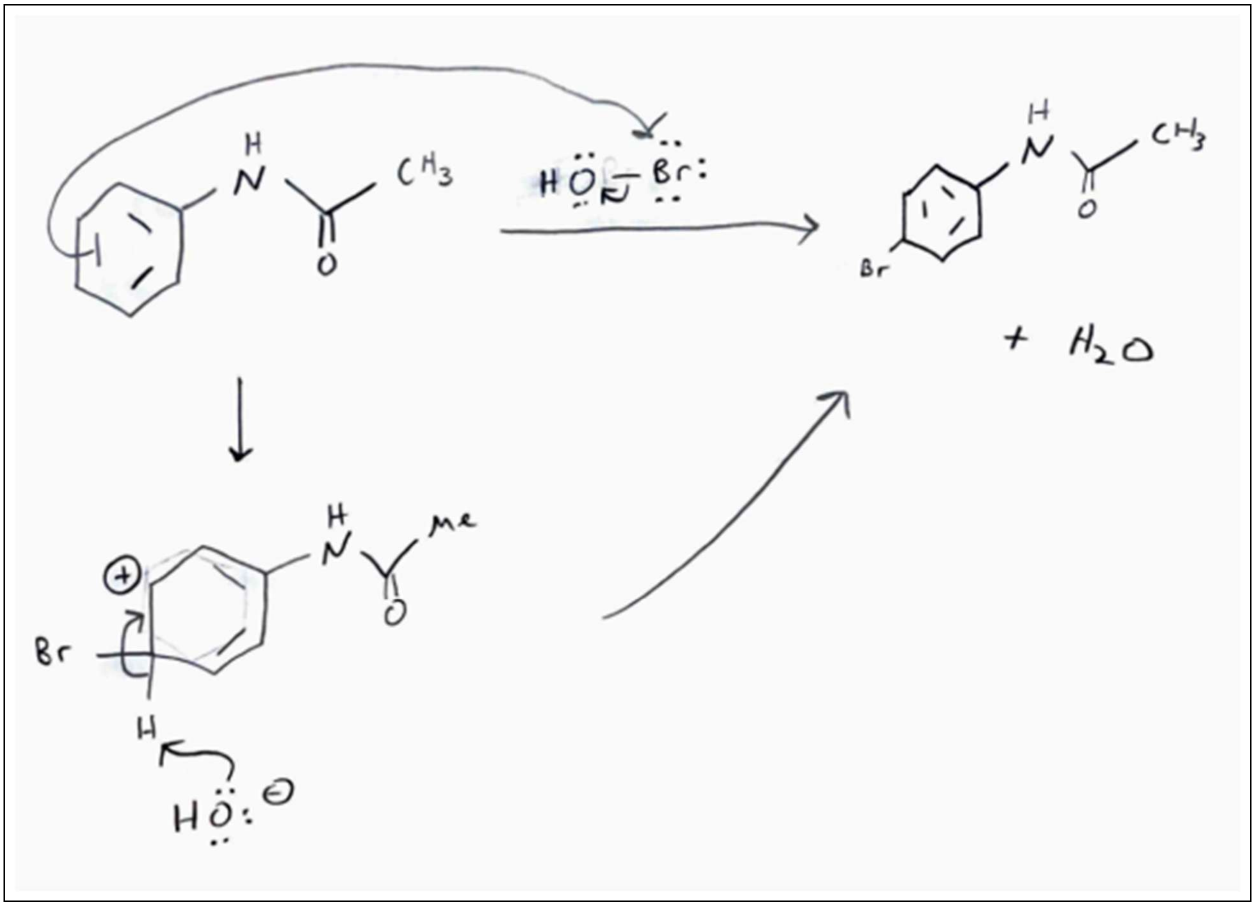

Figure 1: Mechanism for Green Bromination of Acetanilide

Procedure

The procedures of this experiment are as follows:

Materials

A list of utilized chemicals:

- 95% Ethanol : 5% Water
- Acetic Acid
- Acetanilide
- Sodium Bromide
- Sodium Hypochlorite
- Sodium Thiosulfate solution
- Sodium Hydroxide

A list of utilized equipment:

- Ice bath
- 125 mL Erlenmeyer flask
- Vacuum filtration setup
- Recrystallization setup
- Melting point apparatus
- IR spectrometer
- H-NMR spectrometer

Equations

❖ $Percent\ Recovery = \left(\frac{Mass\ of\ Crude}{Mass\ of\ Pure}\right) X\ 100$

❖ $Percent\ Yield = \left(\frac{Mass\ of\ Actual\ Yield}{Mass\ of\ Theoretica\ \ Yield}\right) X\ 100$

Methods

The general methodology for this experiment is as follows. First, I set an ice bath and let sit until it reached roughly 5 °C. Then, I placed 6.00 mL of 95% ethanol and 5.00 mL of acetic acid into a 125 mL Erlenmeyer flask with 0.9994g of acetanilide and 1.7946g of sodium bromide. I then swirled the flask until all the reagents became dissolved, and then I set it in the ice bath for 5 minutes to cool down. After, I added 5.60 mL of sodium hypochlorite (household bleach) in the reaction flask and let it sit in the ice bath for another 5 minutes. Then, I allowed the reaction to come to room temperature by taking the flask out of the ice bath and sitting on the table for 15 minutes. Then, I added 5.00 mL of sodium thiosulfate and 5.00 mL of sodium hydroxide solution to quench to bromine and stop the reaction (bromine color disappears). After, I vacuumed the solid material and calculated the crude mass. Finally, I performed a recrystallization of the crude product with 95% ethanol : 5% water to obtain the pure product.

After the dry, pure product was synthesized, I obtained a melting point range using the melting point apparatus, an IR spectrum using the IR spectrometer, and an H-NMR graph using the H-NMR spectrometer.

Results

Table 1: Reagent Data

	Molar Mass (g/mol)	Density (g/mL)	Quantity Used	Moles (mol)
Acetanilide	135.17	N/A	0.9994g	0.007394
Ethanol	46.07	0.789	6.00mL	0.103
Acetic Acid	60.05	1.049	5.00mL	0.0873
Sodium Bromide	102.89	3.20	1.7946g	0.017442
Sodium Hypochlorite	74.44	1.108	5.60mL	0.0834

Calculation 1: Example Math for Calculating Acetanilide Moles

$$\frac{0.9994\ g}{1}\ X\ \frac{1\ mol}{135.17\ g} = 0.007394\ mol$$

Table 2: Percent Recovery

Mass of Crude Product (g)	Mass of Pure Product (g)	Percent Recovery (%)
0.7877	0.1565	19.87

Calculation 2: Example Math for Calculating Percent Recovery

$$\frac{0.1565\ g}{0.7877\ g}\ X\ 100 = 19.87\ \%$$

Table 3: Percent and Theoretical Yield

	Molar Mass (g/mol)	Quantity Obtained (g)	Theoretical Yield (g)	Percent Yield (%)
p-bromoacetanilide	214.06	0.1565	1.583	9.886

Calculation 3: Example Math for Percent Yield

$$\frac{0.1565\ \text{g}}{1.583\ \text{g}} \text{ X } 100 = 9.886\ \%$$

Table 4: Experimental and Literature Melting Point of p-Bromoacetanilide

	Melting Point (low, °C)	Melting Point (high, °C)
Experimental	163	166
Literature	165	168

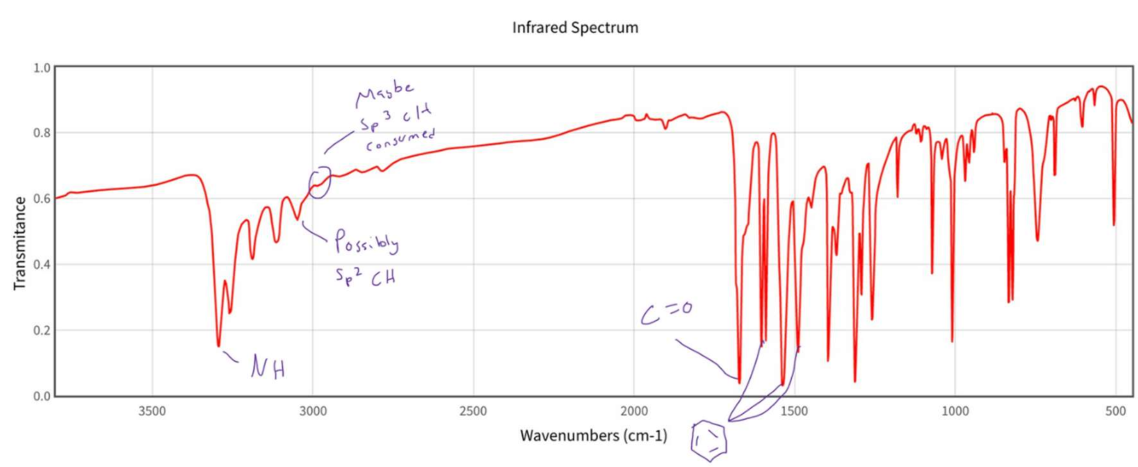

Figure 2: Annotated Infrared Spectroscopy of p-Bromoacetanilide

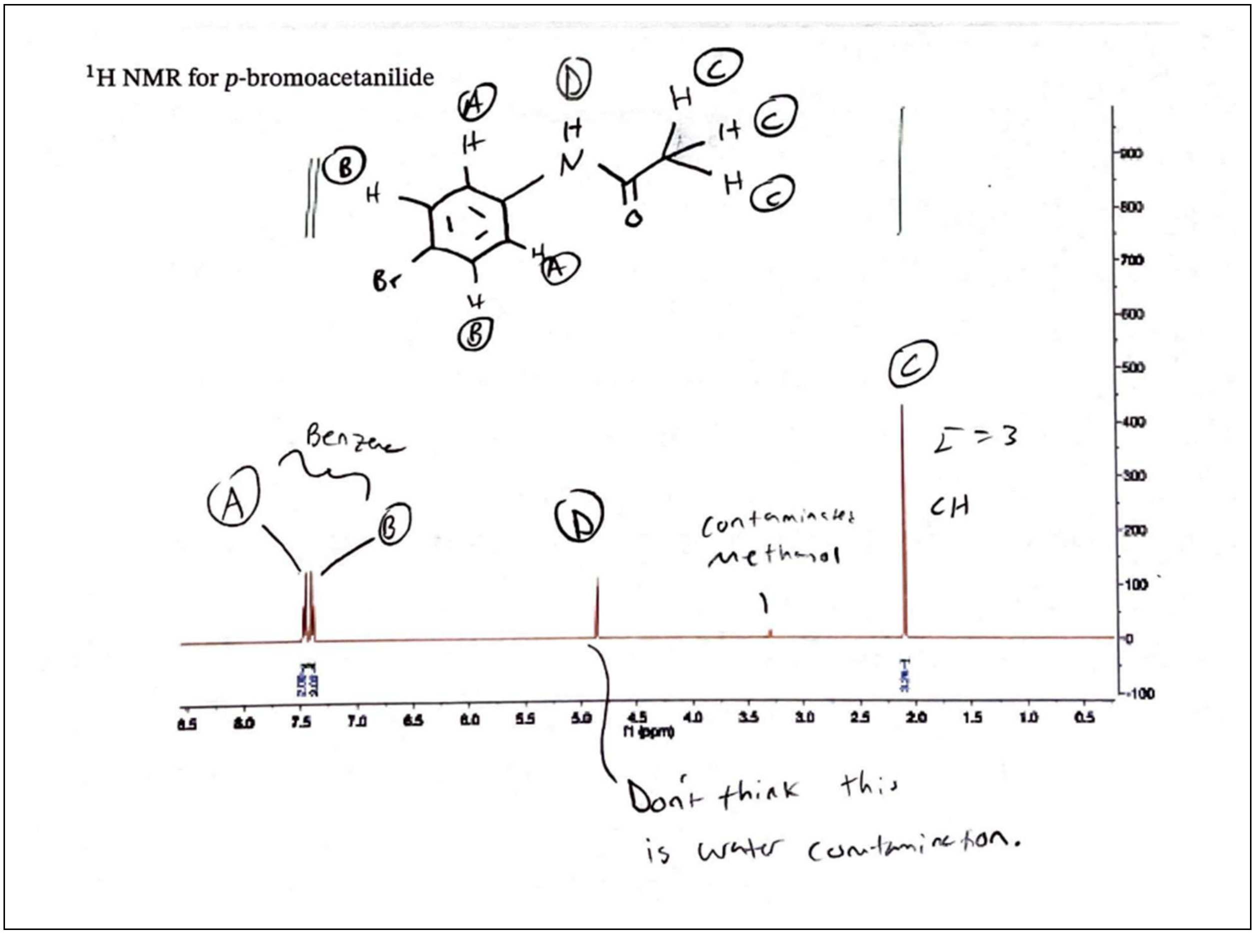

Figure 3: Annotated H-NMR Spectroscopy of p-Bromoacetanilide

Discussion

This discussion demonstrates the significance of the results and important findings. The experiment produced 0.7877g of crude solid, which was recrystallized to 0.1565g of pure product. Utilizing acetanilide as the limiting reagent (0.00734 mol), the theoretical yield of p-bromoacetanilide was 1.583g, making the final percent yield to be 9.886%, which is much lower than expected for this reaction (*Table 3*).

Despite generating a low yield, the melting point range (163 – 166 °C) was close to the literature range (165 – 168 °C), indicating a high purity of my final product (*Table 4*). Also, my melting point range had a difference of 3 across both extrema, which is the same difference for the literature range, signifying a narrow range and high purity of my final product. However, there might be slight impurities in my sample due to the slightly depressed melting point range. Overall, the melting point range and narrow difference agreed with my initial expectations and hypothesis.

In addition, the infrared spectrometer revealed that my final product was fairly pure as all relative peaks associated with p-bromoacetanilide were present. For example, the amide C=O stretch at 1650 cm^{-1} and N-H stretch in the 3300 cm^{-1} region alongside the benzene ring peaks (1450 – 1600 cm^{-1}) are all indicative of a p-bromoacetanilide sample compared to acetanilide contamination (*Figure 2*). Overall, the infrared spectra data agreed with my expectations and hypothesis to receive accurate, graphical results.

Furthermore, the H-NMR spectrum displayed many peaks that are highly consistent with p-bromoacetanilide. First, two peaks were present around 7.5 ppm indicating the two sets of hydrogens around the benzene ring (*Figure 3*). Additionally, those two peaks indicate that this sample was not o-bromoacetanilide (the ortho substituted version) because two even identical peaks around the same shift value would not be present for the hydrogens attached to the benzene ring due to asymmetry. To continue, peaks D and C are consistent with the amide portion of p-bromoacetanilide, even though these peaks would also be present in acetanilide. (*Figure 3*). Overall, the H-NMR spectrum agreed with my hypothesis and initial expectations to retrieve appropriate and accurate results.

In regard to possible errors, there could have been several reasons to account for the unexpected or slightly skewed results. To start, only a 19.87% recovery of sample was retrieved after recrystallization, so it is possible that too much hot ethanol/water solvent was used, leaving a significant amount of p-bromoacetanilide dissolved and unable to precipitate out (*Table 2*). Further, transferring materials back and forth through vacuum filtration can lead to a loss of material; thus, contributing to the lower percent yield alongside overuse of solvent in recrystallization. Finally, the quenching reagents could have been introduced too early, stopping the reaction before total completion.

Note: All calculation examples are related to the table above it.

Conclusion

In summary, this experiment has succeeded in demonstrating the green bromination of acetanilide to p-bromoacetanilide, the para directed product. Additionally, this experiment supported my hypothesis that the H-NMR and IR spectrum would be fairly accurate, and the melting point range will be narrow and within a close range. However, my expectation of a high percentage yield of final product was contradicted. Regarding possible errors, the results seem to conclude from human error, such as too much hot recrystallization solvent used to purify crude product, loss of material through multiple vacuum transfers, and possibly quenching the reaction too early, hindering further production of final product. Finally, for the future, I would suggest that a more careful recrystallization process be taken to better receive a greater percent yield of product, especially if the pure sample is needed for later reactions or multipart synthesis.